

# Learning Koopman Operators

①

Given Koopman operator  $K: L^2(\Omega) \rightarrow L^2(\Omega)$ ,  
and a dictionary of observables  $\{\phi_1, \dots, \phi_N\} \subset L^2(\Omega)$

GOAL: Learn approx.  $\hat{K}_N: V_N \rightarrow V_N$ , where

$$V_N = \text{span}\{\phi_1, \dots, \phi_N\} \subset L^2(\Omega).$$

Galerkin Approx. Minimize the residuals

$$\rho_i(x) = [K\phi_i](x) - \sum_{j=1}^N [K_N]_{ji} \phi_j(x)$$

①  $\Rightarrow$  The "best" approximation takes the form

$$\hat{K}_N = \underset{K_N \in \mathbb{C}^{N \times N}}{\operatorname{argmin}} \left\| K[\phi_1 \dots \phi_N] - [\phi_1 \dots \phi_N] K_N \right\|$$

$j^{\text{th}} \text{ col. is } \rho_j(x)$

where  $\|[\rho_1 \dots \rho_N]\| = \sum_{j=1}^N \int_{\Omega} |\rho_j(x)|^2 dx$ .

③  $\Rightarrow$  Analogous to DMD but  $K_N$  appears on right.

①  $\Rightarrow \hat{K}_N$  satisfies the linear system of eqn's:

$$\underbrace{(\rho_i \perp V_N)}_{i=1, \dots, N}$$

$$\sum_{i=1}^N \langle \phi_i, \phi_j \rangle [\hat{K}_N]_{ji} = \langle K\phi_i, \phi_j \rangle$$

## Quadrature Approximation

In practice, we do not have observe  $\phi_j(x)$  and  $[K\phi_j](x) = \phi_j(f(x))$  at every point  $x \in \Omega$ , and we cannot compute  $\langle \phi_i, \phi_j \rangle$  and  $\langle \phi_j \circ f, \phi_j \rangle$  directly. Instead, we observe the pairs

$$\phi_j(x_k) \text{ and } \phi_j(y_k), \quad k=1, \dots, m.$$

Q: How can we approximate the required inner products  $\langle \phi_i, \phi_j \rangle$  and  $\langle \phi_j \circ f, \phi_j \rangle$  from the snapshot data  $X, Y$ ?

Idea: Approximate  $L^2$  inner products w/ quadrature

$$\begin{aligned} \langle f, g \rangle &= \int_{\Omega} f(x) \overline{g(x)} dx \\ &\approx \sum_{k=1}^m w_k f(x_k) \overline{g(x_k)} \end{aligned}$$

Here,  $\{w_k\}_{k=1}^m = \text{quad. weights}$

$\{x_k\}_{k=1}^m = \text{quad. nodes (= input snap. data)}$

# Weighted Least-Squares

(3) (4)

Applying our quadrature rule, we get

$$\langle \phi_i, \phi_j \rangle \approx \sum_{k=1}^m w_k \phi_i(x_k) \overline{\phi_j(x_k)}$$

$$\begin{aligned} \langle \phi_i \circ F, \phi_j \rangle &\approx \sum_{k=1}^m w_k \phi_i(F(x_k)) \overline{\phi_j(x_k)} \\ &= \sum_{k=1}^m w_k \phi_i(y_k) \overline{\phi_j(x_k)} \end{aligned}$$

For practical computation & analysis, it is useful to collect our results in matrix form.

$$\Phi_X^N = \begin{bmatrix} \phi_1(x_1) & \dots & \phi_N(x_1) \\ \vdots & & \vdots \\ \phi_1(x_m) & \dots & \phi_N(x_m) \end{bmatrix}, \quad \Phi_Y^N = \begin{bmatrix} \phi_1(y_1) & \dots & \phi_N(y_1) \\ \vdots & & \vdots \\ \phi_1(y_m) & \dots & \phi_N(y_m) \end{bmatrix}$$

Let  $W = \text{diag}(w_1, \dots, w_m)$ , and calculate

$$\langle \phi_i, \phi_j \rangle \approx [\Phi_X^{N*} W \Phi_X^N]_{ji} = G^m$$

$$\langle \phi_i \circ F, \phi_j \rangle \approx [\Phi_X^{N*} W \Phi_Y^N]_{ji} = A^m$$

Normal  
Eqn. for  
"Weighted  
Least-squares"  
 $\Rightarrow$

$$G \hat{K}_N^m = A^m$$

## "Data-Driven" Residual

(4) (8)

This weighted least-squares problem implies

$$\Phi_x^* W (\Phi_x \hat{K}_N^m - \Phi_y) = 0,$$

which is equivalent to minimizing

$$\begin{aligned} \|W^{1/2} R\|_F^2 &= \sum_{j=1}^N \sum_{k=1}^m w_k |\rho_j(x_k)|^2 \\ &= \sum_{j=1}^N w_j \sum_{k=1}^m w_k |\phi_j(x_k) - \sum_{i=1}^N [\hat{K}_N^m]_{ij} \phi_i(x_k)|^2 \end{aligned}$$

I.e., minimizing the residual on the data with respect to the weighted Frob. norm.

### Convergence (quadrature)

⊗ If the quadrature rule converges on  $V_N$ ,

$$\int_{\Omega} f(x) dx = \lim_{m \rightarrow \infty} \sum_{k=1}^m w_k f(x_k), \quad \forall f \in V_N,$$

then  $\lim_{m \rightarrow \infty} \hat{K}_N^m = \hat{K}_N$ , the Galerkin approx. of  $K$ .

Q: What can we say about the Galerkin Approx?



## Convergence (Galerkin/Pichler)

(5)

Similar to HW 2, one can develop data-driven a posteriori error bounds for the Galerkin approx.

$\Rightarrow$  See Section 3 of "Intro. Guide Koop. Learn."

Suppose that I have  $g \in L^2(\Omega)$  and  $V_N$  s.t.

$$g_* = \underset{\substack{f \\ f \in V_N}}{\operatorname{argmin}} \|g - f\|_{L^2(\Omega)},$$

and that  $\varepsilon > 0$  is s.t.  $\|g - g_*\|_{L^2(\Omega)} < \varepsilon$ . <sup>set</sup> Then

$$g_* = c_1 \phi_1 + \dots + c_N \phi_N = \left[ \phi_1 \dots \phi_N \right] \varepsilon,$$

so that  ~~$Kg \approx$~~   $g \approx \Phi \varepsilon$  and  $Kg \approx \Phi \hat{K}_N \varepsilon$ .

$$\begin{aligned} \|Kg - \Phi \hat{K}_N \varepsilon\| &\leq \|Kg - \Phi K \Phi \varepsilon\| + \|\Phi K \Phi \varepsilon - \Phi \hat{K}_N \varepsilon\| \\ &\leq \|K\| \|g - g_*\| + \|\Phi K \Phi \varepsilon - \Phi \hat{K}_N \varepsilon\| \\ &\leq \varepsilon \|K\| + \left[ \sum_{j=1}^N \|\rho_j\|_{L^2(\Omega)}^2 \right] \|\varepsilon\| \end{aligned}$$

$\Rightarrow$  Can estimate residuals from data and  $\|K\|$  often known.

$\Rightarrow$  Typically  $\Phi \hat{K}_N \varepsilon \rightarrow Kg$  as  $N \rightarrow \infty$ , <sup>(strong conv.)</sup> but this

$\Rightarrow$  Can extend to powers of  $K$  (prediction).